AAMEENA I(192511284)

Experiment 1

Simple calculator (add, subtract, multiply, divide)

shalini ai.py - C:\Users\shali\OneDrive\Desktop\shalini ai.py (3.5.3)

File Edit Format Run Options Window Help

```
def add(x, y):
  return x + y
def subtract(x, y):
  return x - y
def multiply(x, y):
  return x * y
def divide(x, y):
  return x / y
print("Select operation.")
print("1.Add")
print("2.Subtract")
print("3.Multiply")
print("4.Divide")
choice = input("Enter choice(1/2/3/4): ")
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
if choice == '1':
    print(num1,"+",num2,"=", add(num1,num2))
elif choice == '2':
    print(num1,"-",num2,"=", subtract(num1,num2))
elif choice == '3':
    print(num1,"*",num2,"=", multiply(num1,num2))
elif choice == '4':
    print(num1,"/",num2,"=", divide(num1,num2))
else:
    print("Invalid input")
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, J
an 16 2017, 15:51:26) [MSC v.1900 32
bit (Intel)] on win32
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nse()" for more information.
>>>
========== RESTART: C:\Users\shali\
OneDrive\Desktop\shalini ai.py =====
======
Select operation.
1.Add
2.Subtract
3.Multiply
4.Divide
Enter choice(1/2/3/4): 3
Enter first number: 23
Enter second number: 43
23.0 * 43.0 = 989.0
>>>
```

Experiment 2

Addition of two matrices

shalini ai2.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini ai2.py (3.5.3)

File Edit Format Run Options Window Help

```
matrix1 = [[1, 2, 3],
[4, 5, 6],
[7, 8, 9]]
matrix2 = [[10, 11, 12],
[13, 14, 15],
[16, 17, 18]]
result = [[0, 0, 0],
[0, 0, 0],
[0, 0, 0]]
for i in range(len(matrix1)):
 for j in range(len(matrix1[0])):
   # Add corresponding elements of matrices
   result[i][j] = matrix1[i][j] + matrix2[i][j]
for row in result:
 print(row)
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 16 2017, 15:51:
26) [MSC v.1900 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more in
formation.
>>>
= RESTART: C:\Users\shali\OneDrive\Desktop\shalini pyt
hon ai\shalini ai2.py =
[11, 13, 15]
[17, 19, 21]
[23, 25, 27]
>>>
```

Experiment 3

Transpose of a matrix

shalini ai3.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini ai3.py (3.5.3)

File Edit Format Run Options Window Help

```
matrix = [[1, 2, 3],
[4, 5, 6],
[7, 8, 9]]
rows = len(matrix)
cols = len(matrix[0])
transpose_matrix = [[0 for x in range(rows)] for y in range(cols)]
for i in range(rows):
  for j in range(cols):
    transpose_matrix[j][i] = matrix[i][j]
print("Original Matrix:")
for row in matrix:
  print(row)
print("Transposed Matrix:")
for row in transpose_matrix:
  print(row)
```

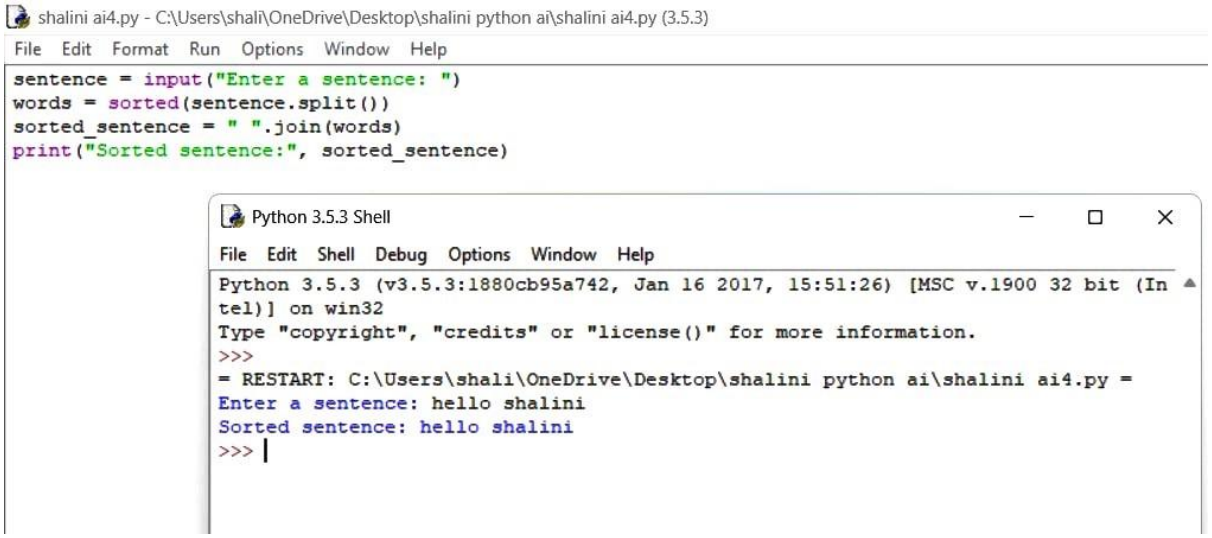
Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 1
6 2017, 15:51:26) [MSC v.1900 32 bit (In
tel)] on win32
Type "copyright", "credits" or "license(
)" for more information.
>>>
= RESTART: C:\Users\shali\OneDrive\Deskt
op\shalini python ai\shalini ai3.py =
Original Matrix:
[1, 2, 3]
[4, 5, 6]
[7, 8, 9]
Transposed Matrix:
[1, 4, 7]
[2, 5, 8]
[3, 6, 9]
>>>
```

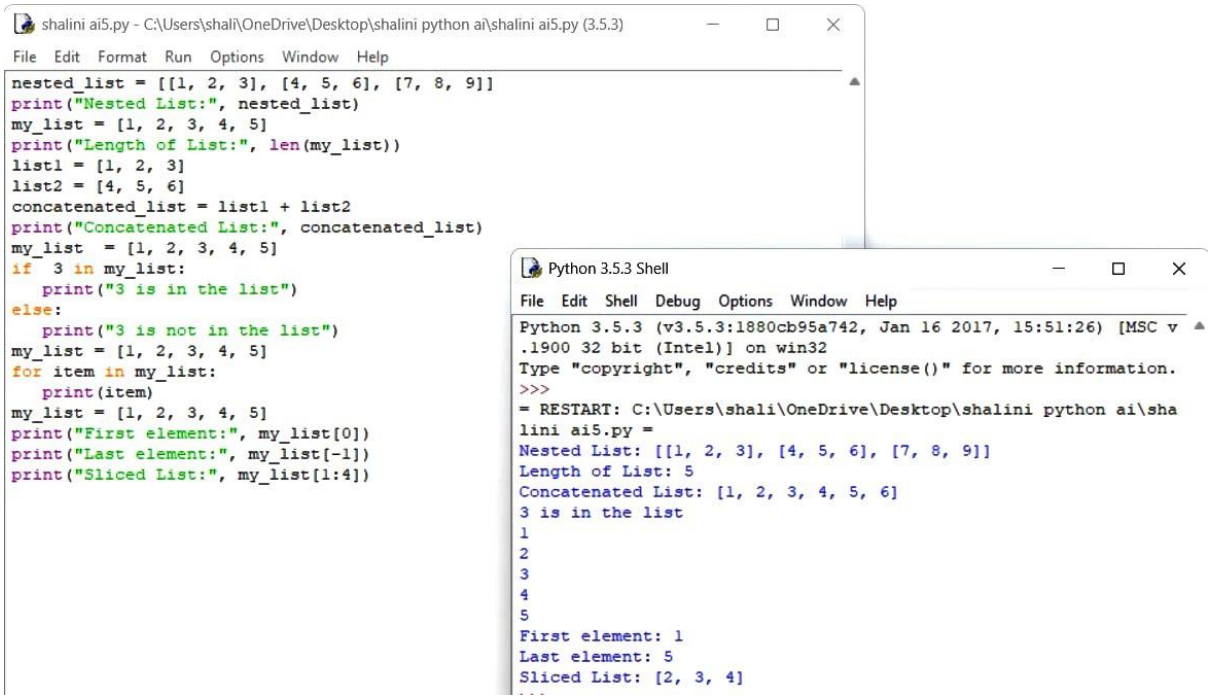
Experiment 4

Sort the words of a sentence alphabetically



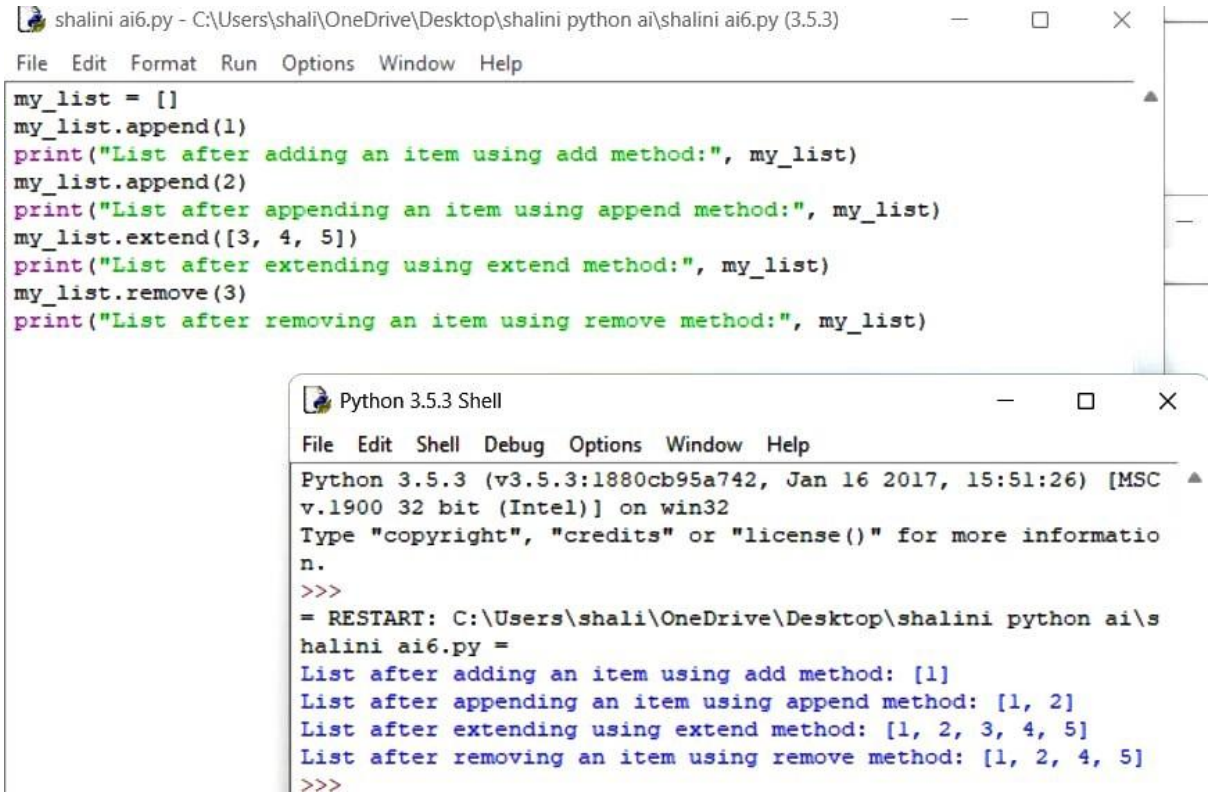
Experiment 5

List operations (nested list, length, concatenation, membership, indexing, slicing)



Experiment 6

List methods (append, extend, remove)



Experiment 7

Set operations (union, intersection, difference, subset)

shalini ai7.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini ai7.py (3.5.3)

File Edit Format Run Options Window Help

```
set1 = {1, 2, 3, 4, 5}
set2 = {4, 5, 6, 7, 8}
print("Set 1:", set1)
print("Set 2:", set2)
intersection_set=set1.intersection(set2)
print("Intersection of set1 and set2:", intersection_set)
union_set = set1.union(set2)
print("Union of set1 and set2:", union_set)
difference_set = set1.difference(set2)
print("Difference between set1 and set2:", difference_set)
symmetric_difference_set = set1.symmetric_difference(set2)
print("Symmetric difference between set1 and set2:", symmetric_difference_set)
subset_check = set1.issubset(set2)
print("Set 1 is subset of set 2:", subset_check)
```

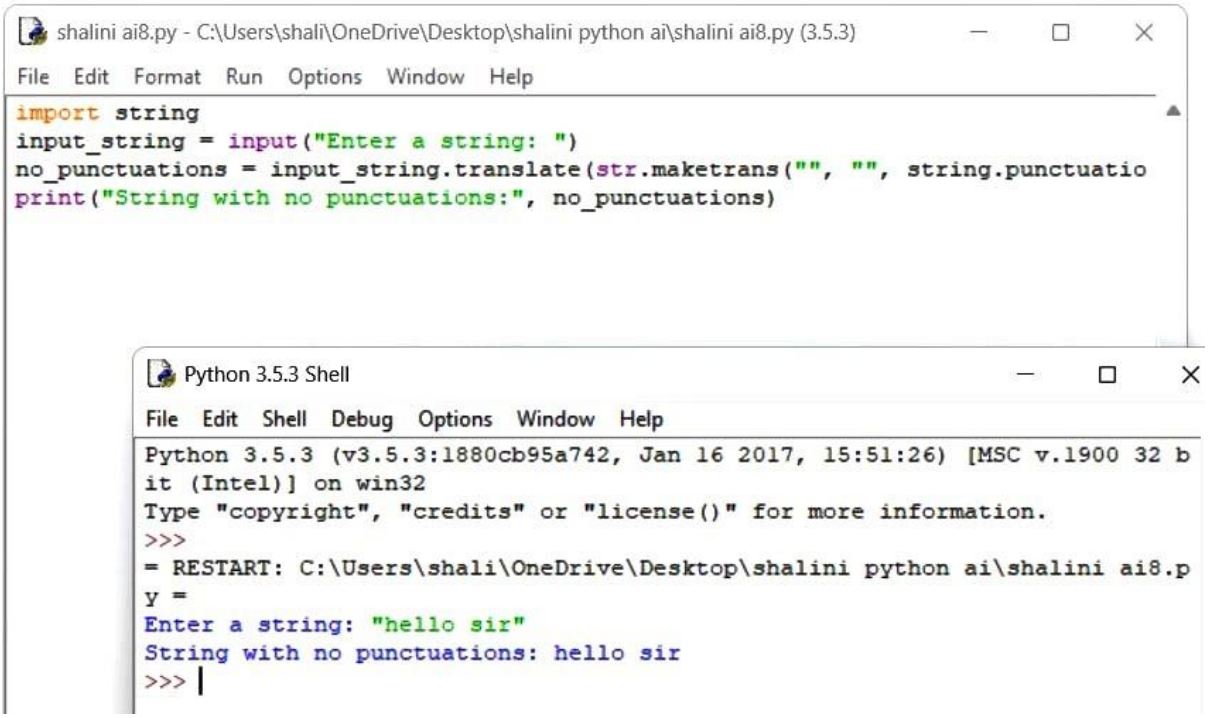
Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 16 2017, 15:51:26) [MSC v.1900 32
t (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini ai7
=
Set 1: {1, 2, 3, 4, 5}
Set 2: {8, 4, 5, 6, 7}
Intersection of set1 and set2: {4, 5}
Union of set1 and set2: {1, 2, 3, 4, 5, 6, 7, 8}
Difference between set1 and set2: {1, 2, 3}
Symmetric difference between set1 and set2: {1, 2, 3, 6, 7, 8}
Set 1 is subset of set 2: False
>>>
```

Experiment 8

Remove punctuation from a string



Experiment 9

Display the calendar of a given month

shalini ai9.py - C:/Users/shali/OneDrive/Desktop/shalini python ai/shalini ai9.py (3.5.3)

File Edit Format Run Options Window Help

```
import calendar

year = int(input("Enter year: "))
month = int(input("Enter month: "))

print(calendar.month(year, month))
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 16 2017, 15:51:26) [MSC v.190
0 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/shali/OneDrive/Desktop/shalini python ai/shalini
ai9.py =
Enter year: 2009
Enter month: 04
     April 2009
Mo Tu We Th Fr Sa Su
       1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30
```

Experiment 10

8-Puzzle problem using A* search

shalini ai10.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini ai10.py (3.5.3)

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```
from queue import PriorityQueue

goal_state = (1, 2, 3, 4, 5, 6, 7, 8, 0)

def heuristic(state):
    distance = 0  # FIX: Initialize distance to 0
    for i in range(3):
        for j in range(3):
            tile = state[i * 3 + j]
            if tile != 0:
                x, y = (tile - 1) // 3, (tile - 1) % 3
                distance += abs(x - i) + abs(y - j)
    return distance

def solve(initial_state):
    frontier = PriorityQueue()
    # Store elements as (f_score, g_score, state)
    # initial state has g_score = 0
    frontier.put((heuristic(initial_state), 0, initial_state))
    explored = set()

    while not frontier.empty():
        _, g, state = frontier.get()  # FIX: Track the accumulated cost 'g'

        if state == goal_state:
            return True

        if state in explored:
            continue
        explored.add(state)

        for successor in successors(state):
            if successor not in explored:
                g_successor = g + 1  # FIX: Increment path cost by 1 from the parent
                f_successor = g_successor + heuristic(successor)
                frontier.put((f_successor, g_successor, successor))

    return False

def successors(state):
    successors = []
    i = state.index(0)

    if i % 3 != 0:  # Slide tile to the left
        new_state = list(state)
        new_state[i], new_state[i - 1] = new_state[i - 1], new_state[i]
        successors.append(tuple(new_state))
```

Experiment 10 (continued)

```
    if i % 3 != 0:  # Slide tile to the left
        new_state = list(state)
        new_state[i], new_state[i - 1] = new_state[i - 1], new_state[i]
        successors.append(tuple(new_state))

    if i % 3 != 2:  # Slide tile to the right
        new_state = list(state)
        new_state[i], new_state[i + 1] = new_state[i + 1], new_state[i]
        successors.append(tuple(new_state))

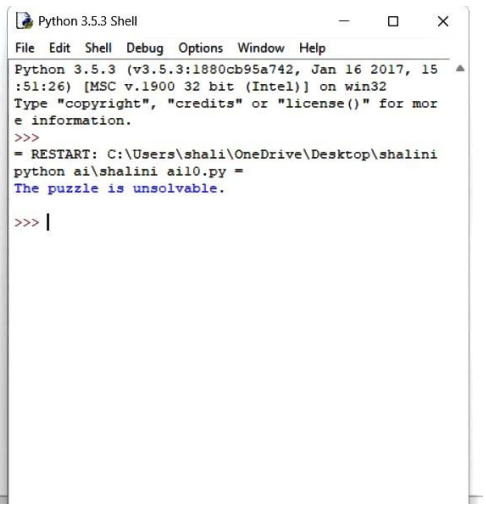
    if i // 3 != 0:  # Slide tile up
        new_state = list(state)
        new_state[i], new_state[i - 3] = new_state[i - 3], new_state[i]
        successors.append(tuple(new_state))

    if i // 3 != 2:  # Slide tile down
        new_state = list(state)
        new_state[i], new_state[i + 3] = new_state[i + 3], new_state[i]
        successors.append(tuple(new_state))

    return successors

initial_state = (2, 8, 3, 1, 6, 4, 7, 0, 5)

if solve(initial_state):
    print("The puzzle is solvable!")
else:
    print("The puzzle is unsolvable.")
```



Experiment 11

8-Queens problem using backtracking

shalini ai11.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini ai11.py (3.5.3)

File Edit Format Run Options Window Help

```
def solve_queens(n):
    board = [-1] * n
    result = []

    def is_valid(row, col):
        for i in range(row):
            if board[i] == col or abs(row - i) == abs(col - board[i]):
                return False
        return True

    def backtrack(row):
        if row == n:
            result.append(list(board))
            return

        for col in range(n):
            if is_valid(row, col):
                board[row] = col
                backtrack(row + 1)
                board[row] = -1

    backtrack(0)
    return result


solutions = solve_queens(8)

for solution in solutions:
    for row in range(8):
        line = ""
        for col in range(8):
            if solution[row] == col:
                line += "Q "
            else:
                line += "- "
        print(line)
    print("\n")
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 16 2017, 15:51:26) [MSC v.1
900 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shali\OneDrive\Desktop\shalini python ai\shali
ni aill.py =
Q - - - - - - -
- - - - Q - - -
- - - - - - - Q
- - - - - Q - -
- - Q - - - - -
- - - - - - Q -
- Q - - - - - -
- - - Q - - - -
```

Experiment 12

Water jug problem using breadth-first search

shalini12.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini12.py (3.5.3)

File Edit Format Run Options Window Help

```
from collections import deque

def water_jug_problem(x, y, z):
    """
    Solves the water jug problem using breadth-first search.
    x: size of jug 1
    y: size of jug 2
    z: target amount of water
    Returns the sequence of steps to reach the target amount of water.
    """
    queue = deque([(0, 0, [])])  # Start with both jugs empty
    visited = set()

    while queue:
        a, b, steps = queue.popleft()

        if (a, b) in visited:
            continue

        visited.add((a, b))

        if a == z or b == z:
            return steps

        queue.append((x, b, steps + ['fill jug 1']))
        queue.append((a, y, steps + ['fill jug 2']))
        queue.append((0, b, steps + ['empty jug 1']))
        queue.append((a, 0, steps + ['empty jug 2']))

        amount = min(a, y - b)
        queue.append((a - amount, b + amount, steps + ['pour jug 1 into jug 2']))

        # Pour jug 2 into jug 1
        amount = min(x - a, b)
        queue.append((a + amount, b - amount, steps + ['pour jug 2 into jug 1']))

    return None


steps = water_jug_problem(4, 3, 2)

if steps:
    print('\n'.join(steps))
else:
    print('No solution found.')
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 16
2017, 15:51:26) [MSC v.1900 32 bit (Intel)
] on win32
Type "copyright", "credits" or "license()"
for more information.
>>>
== RESTART: C:\Users\shali\OneDrive\Deskto
p\shalini python ai\shalini12.py ==
fill jug 2
pour jug 2 into jug 1
fill jug 2
pour jug 2 into jug 1
>>>
```

Ln: 9 Col: 4

Experiment 13

Cryptarithmetic puzzle (SEND + MORE = MONEY)

shalinia.i13.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalinia.i13.py (3.5.3)

File Edit Format Run Options Window Help

```
from itertools import permutations

def solve():
    letters = 'SENDMORY'
    for p in permutations(range(10), 8):
        d = dict(zip(letters, p))
        if d['S'] == 0 or d['M'] == 0:
            continue

        SEND = d['S']*1000 + d['E']*100 + d['N']*10 + d['D']
        MORE = d['M']*1000 + d['O']*100 + d['R']*10 + d['E']
        MONEY = d['M']*10000 + d['O']*1000 + d['N']*100 + d['E']*10 + d['Y']

        if SEND + MORE == MONEY:
            print("SEND =", SEND)
            print("MORE =", MORE)
            print("MONEY =", MONEY)
            break

solve()
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 1
6 2017, 15:51:26) [MSC v.1900 32 bit (In
tel)] on win32
Type "copyright", "credits" or "license(
)" for more information.
>>>
= RESTART: C:\Users\shali\OneDrive\Deskt
op\shalini python ai\shalinia.il3.py =
SEND = 9567
MORE = 1085
MONEY = 10652
>>>
```

Experiment 14

Missionaries and cannibals problem using BFS

shalinia.i14.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalinia.i14.py (3.5.3)

File Edit Format Run Options Window Help

```
from collections import deque

def solve():
    q = deque([((3, 3, 1), [])])
    visited = set()

    while q:
        (m, c, b), path = q.popleft()

        if (m, c, b) == (0, 0, 0):
            return path + [(m, c, b)]

        if (m, c, b) in visited:
            continue
        visited.add((m, c, b))

        for x, y in [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]:
            nm = m - x if b else m + x
            nc = c - y if b else c + y

            if 0 <= nm <= 3 and 0 <= nc <= 3:
                if (nm == 0 or nm >= nc) and (3-nm == 0 or 3-nm >= 3-nc):
                    q.append(((nm, nc, 1-b), path + [(m, c, b)]))

    return []

for state in solve():
    print(state)
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 16 2017,
15:51:26) [MSC v.1900 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for m
ore information.
>>>
= RESTART: C:\Users\shali\OneDrive\Desktop\shali
ni python ai\shalinia.il4.py =
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
>>>
```

Experiment 15

Vacuum cleaner agent

shalinia.i15.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalinia.i15.py (3.5.3)

File Edit Format Run Options Window Help

```
def vacuum_cleaner(room):
    for i in range(len(room)):
        if room[i] == 1:
            print("Room", i, "is dirty. Cleaning...")
            room[i] = 0
        else:
            print("Room", i, "is clean.")

    print("All rooms are clean!")

room = [1, 0]  # 1 = Dirty, 0 = Clean
vacuum_cleaner(room)
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 16 2017, 15:51:26) [MSC v.19
00 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shali\OneDrive\Desktop\shalini python ai\shalin
ia.i15.py =
Room 0 is dirty. Cleaning...
Room 1 is clean.
All rooms are clean!
```

Experiment 16

Breadth-first search (BFS) graph traversal

shalini ai16.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini ai16.py (3.5.3)

File Edit Format Run Options Window Help

```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])

    while queue:
        node = queue.popleft()

        if node not in visited:
            print(node, end=" ")
            visited.add(node)

            for neighbour in graph[node]:
                if neighbour not in visited:
                    queue.append(neighbour)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

print("BFS Traversal:")
bfs(graph, 'A')
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 16 2017, 15:51:26) [MSC v.1900 32
 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini ail
6.py =
BFS Traversal:
A B C D E F
>>>
```

Experiment 17

Depth-first search (DFS) graph traversal

shalini ai17.py - C:\Users\shali\OneDrive\Desktop\shalini python ai\shalini ai17.py (3.5.3)

File Edit Format Run Options Window Help

```
def dfs(graph, node, visited):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbour in graph[node]:
            dfs(graph, neighbour, visited)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

print("DFS Traversal:")
dfs(graph, 'A', set())
```

Python 3.5.3 Shell

File Edit Shell Debug Options Window Help

```
Python 3.5.3 (v3.5.3:1880cb95a742, Jan 16 2017, 15:51
:26) [MSC v.1900 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more i
nformation.
>>>
= RESTART: C:\Users\shali\OneDrive\Desktop\shalini py
thon ai\shalini ai17.py =
DFS Traversal:
A B D E C F
>>>
```